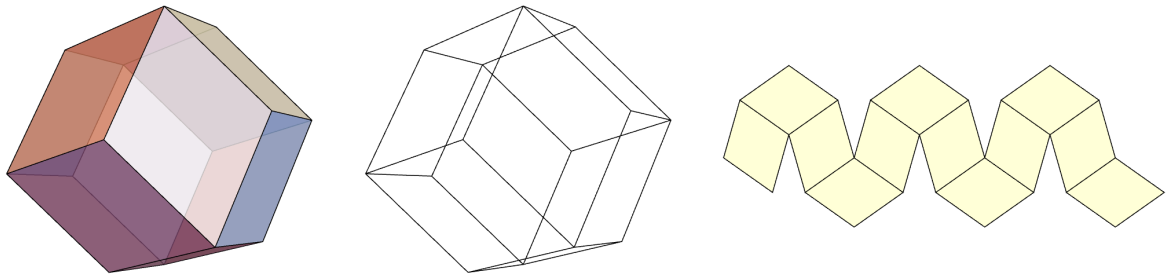
Rhombic Dodecahedron

Download Wolfram Notebook



MAKE YOUR OWN RHOMBIC DODECAHEDRON
Print & fold
Print in 3D

The (first) rhombic dodecahedron is the dual polyhedron of the cuboctahedron (Holden 1971, p. 55). It is sometimes also called the rhomboidal dodecahedron (Cotton 1990), and the "first" may be included when needed to distinguish it from the Bilinski dodecahedron (Bilinski 1960, Chilton and Coxeter 1963). It is illustrated above together with a wireframe version and a net that can be used for its construction.

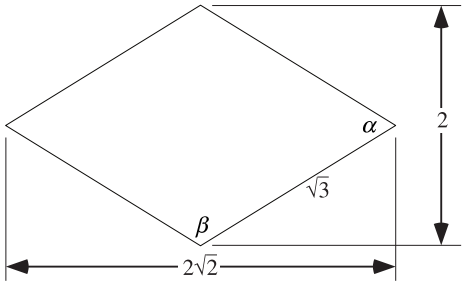
It is Wenninger dual W_{11}.

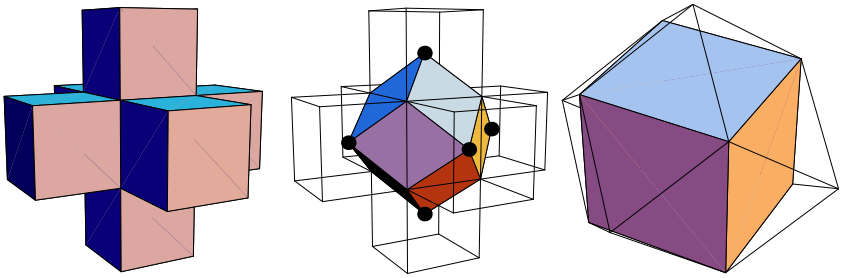
The rhombic dodecahedron is implemented in the Wolfram Language as PolyhedronData["RhombicDodecahedron"].

A rhombic dodecahedron appears in the upper right as one of the polyhedral "stars" in M. C. Escher's woodcut engraving "Stars" (Forty 2003, Plate 43).

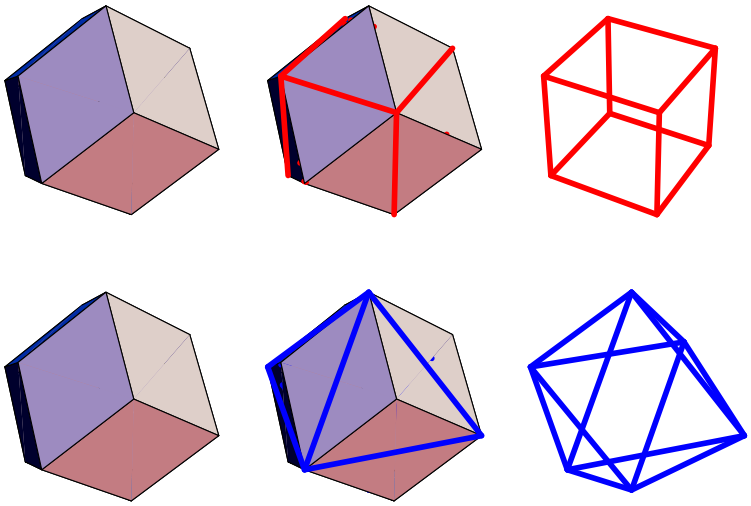
This website uses cookies to optimize your experience with our services on the site, as described in our Privacy Policy.
Accept & Close

$$\alpha = 2\cot^{-1}\sqrt{2} \tag{1}$$
$$= \cos^{-1}\left(\tfrac{1}{3}\right) \tag{2}$$
$$\approx 70.53\,^\circ \tag{3}$$
$$\beta = 2\tan^{-1}\sqrt{2} \tag{4}$$
$$\approx 109.47\,^\circ. \tag{5}$$

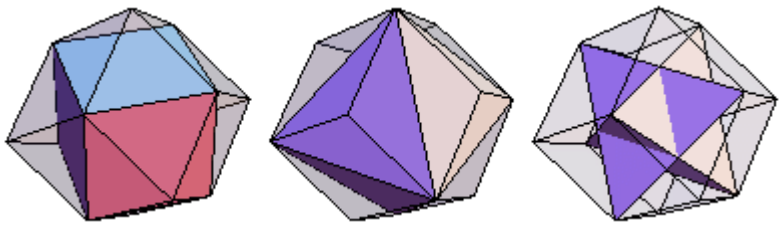




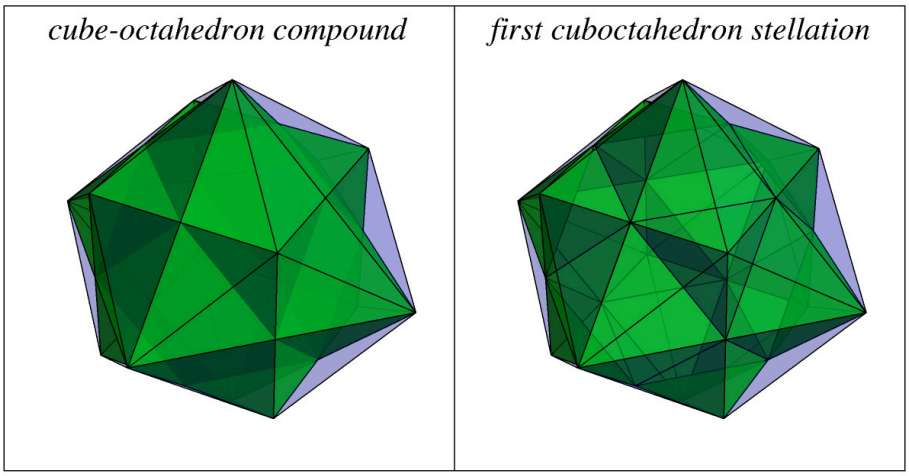
The rhombic dodecahedron can be built up by a placing six cubes on the faces of a seventh, in the configuration of a metal "jack" (left figure). Joining the centers of the outer cubes with the vertices of the central cube then gives the rhombic dodecahedron (middle figure). Affixing a square pyramid of height 1/2 on each face of a cube having unit edge length results in a rhombic dodecahedron (right figure; Brückner 1900, p. 130; Steinhaus 1999, p. 185).



This website uses cookies to optimize your experience with our services on the site, as described in our Privacy Policy.



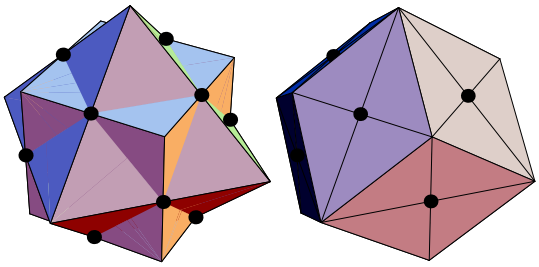
More specifically, a cube, octahedron, and stella octangula can be inscribed in the vertices of a rhombic dodecahedron (E. Weisstein, Dec. 25, 2009).



The rhombic dodecahedron is the convex hull of the cube-octahedron compound and the first cuboctahedron stellation.

If the rhombic dodecahedron is hinged into six square pyramids along three consecutive face diagonals, the resulting model can be folded into a cube (Wells 1991). One possible construction for the rhombic dodecahedron is known as the Bauspiel. It can also be constructed by augmentation of a unit edge-length cube by a pyramid with height 1/2.

The rhombic dodecahedron is a zonohedron and a space-filling polyhedron (Steinhaus 1999, p. 185). The vertices are given by $(\pm 1, \pm 1, \pm 1)$, $(\pm 2, 0, 0)$, $(0, \pm 2, 0)$, $(0, 0, \pm 2)$.



The edges of the cube-octahedron compound intersecting in the points plotted above are the diagonals of rhombi (left figure), and the 12 rhombi form a rhombic dodecahedron

This website uses cookies to optimize your experience with our services on the site, as described in our Privacy Policy.

There are three stellations of the rhombic dodecahedron.

The rhombic dodecahedron can be built using a Haüy construction. The Haüy rhombic dodecahedral numbers

$$\mathrm{HRhoDod}_n = (2n-1)\left(8n^2-14n+7\right) \tag{6}$$

give a method for calculating the volume of the rhombic dodecahedron,

$$V = \lim_{n\to\infty} \mathrm{HRhoDod}_n \left(\frac{a}{n\sqrt{3}}\right)^3 \tag{7}$$

$$= \frac{16}{9}\sqrt{3}\,a^3 \tag{8}$$

(Steinhaus 1999). The surface area of a rhombic dodecahedron with unit edge length is

$$S = 8\sqrt{2}, \tag{9}$$

and its inertia tensor is

$$I = \begin{bmatrix} \frac{1}{3}Ma^2 & 0 & 0 \\ 0 & \frac{1}{3}Ma^2 & 0 \\ 0 & 0 & \frac{1}{3}Ma^2 \end{bmatrix}. \tag{10}$$

SEE ALSO

Bauspiel, Bilinski Dodecahedron, Cube-Octahedron Compound, Dodecahedron, Elongated Dodecahedron, Haüy Construction, Pyritohedron, Rhombic Dodecahedron Stellations, Rhombic Triacontahedron, Rhombus, Sphere Packing, Steinmetz Solid, Trigonal Dodecahedron, Zonohedron

EXPLORE WITH WOLFRAM|ALPHA

WolframAlpha Solve your math problems and get step-by-step solutions

More things to try: = rhombic dodecahedron = 30% of 8 miles = cardioid

REFERENCES

Ball, W. W. R. and Coxeter, H. S. M. *Mathematical Recreations and Essays, 13th ed.* New York: Dover, p. 137, 1987.

Bilinski, S. "Über die Rhombenisoeder." *Glasnik* 15, 251-263, 1960.

This website uses cookies to optimize your experience with our services on the site, as described in our Privacy Policy.

Cotton, F. A. *Chemical Applications of Group Theory, 3rd ed.* New York: Wiley, p. 62, 1990.

Cundy, H. and Rollett, A. "Rhombic Dodecahedron. $V(3.4)^2$." §3.8.1 in *Mathematical Models, 3rd ed.* Stradbroke, England: Tarquin Pub., p. 120, 1989.

Escher, M. C. "Stars." Wood engraving. 1948. http://www.mcescher.com/Gallery/back-bmp/LW359.jpg.

Forty, S. *M.C. Escher.* Cobham, England: TAJ Books, 2003.

Grünbaum, B. "Parallelogram-Faced Isohedra with Edges in Mirror-Planes." *Disc. Math.* **221**, 93-100, 2000.

Holden, A. *Shapes, Space, and Symmetry.* New York: Columbia University Press, p. 55, 1971.

Steinhaus, H. *Mathematical Snapshots, 3rd ed.* New York: Dover, pp. 185-186, 1999.

Wells, D. *The Penguin Dictionary of Curious and Interesting Geometry.* London: Penguin, pp. 215-216, 1991.

Wenninger, M. J. *Dual Models.* Cambridge, England: Cambridge University Press, pp. 19, 21, and 34, 1983.

CITE THIS AS:

Weisstein, Eric W. "Rhombic Dodecahedron." From *MathWorld*--A Wolfram Web Resource. https://mathworld.wolfram.com/RhombicDodecahedron.html

SUBJECT CLASSIFICATIONS

Geometry › Solid Geometry › Polyhedra › Dodecahedra ›

Geometry › Solid Geometry › Polyhedra › Archimedean Duals ›

Geometry › Solid Geometry › Polyhedra › Canonical Polyhedra ›

More...

This website uses cookies to optimize your experience with our services on the site, as described in our Privacy Policy.